AERIS Expert Review Packet

Anomaly

Source / metric	openaq / Ozone (ozone)
Observed / expected baseline	0.075 / 0.0170443 ppm
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-08-02 18:00 UTC
Location	29.5203, -95.3925
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	7 / 704	20.95 to 100; mean 69.3142	42.53 at 2026-08-02 17:55Z; 5 min before event
asos	Air temperature (temperature)	degC	7 / 704	23.8889 to 37; mean 30.2672	34.6111 at 2026-08-02 17:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	7 / 882	0 to 360; mean 179.456	0 at 2026-08-02 17:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	7 / 917	0 to 33.4389; mean 2.906	0 at 2026-08-02 17:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	8 / 104	5876.33 to 5946.41; mean 5902.25	5888.01 at 2026-08-02 18:00Z; at event time
noaa_gfs	Planetary boundary layer height (pbl_height)	m	8 / 104	41.0403 to 2539.72; mean 808.846	2407.74 at 2026-08-02 18:00Z; at event time
noaa_gfs	Precipitable water (precipitable_water)	mm	8 / 104	27.6341 to 57.7464; mean 39.9068	34.901 at 2026-08-02 18:00Z; at event time
noaa_gfs	Surface pressure (surface_pressure)	hPa	8 / 104	1004.02 to 1013.29; mean 1008.98	1008.5 at 2026-08-02 18:00Z; at event time
noaa_gfs	850-hPa temperature (t_850)	degC	8 / 104	18.4016 to 22.5646; mean 20.262	19.9148 at 2026-08-02 18:00Z; at event time
noaa_gfs	10-m eastward wind (u_10m)	m/s	8 / 104	-3.57184 to 4.5116; mean 1.0335	-1.15038 at 2026-08-02 18:00Z; at event time
noaa_gfs	10-m northward wind (v_10m)	m/s	8 / 104	-3.70918 to 6.32564; mean 1.2366	-2.76918 at 2026-08-02 18:00Z; at event time
openaq	Ozone (ozone)	ppm	16 / 1039	0 to 0.081; mean 0.029	0.075 at 2026-08-02 18:00Z; at event time
openaq	PM10 (pm10)	ug/m3	3 / 217	11 to 153; mean 28.4009	17 at 2026-08-02 18:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM2.5 (pm25)	ug/m3	68 / 2670	0 to 50.8; mean 8.2751	17.2 at 2026-08-02 18:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	4 / 288	0 to 100; mean 28.3333	96 at 2026-08-02 18:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	4 / 288	34 to 96; mean 70.7396	47 at 2026-08-02 18:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	4 / 288	0 to 0.22; mean 0.0012	0 at 2026-08-02 18:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	4 / 288	1007 to 1014; mean 1010.8	1011 at 2026-08-02 18:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	4 / 288	25.3 to 37.65; mean 31.149	35.32 at 2026-08-02 18:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	4 / 288	0 to 355; mean 193.833	343 at 2026-08-02 18:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	4 / 288	0.26 to 5.85; mean 2.1803	3.11 at 2026-08-02 18:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	70 / 4934	0 to 313; mean 10.5843	12.5 at 2026-08-02 18:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	8 / 8	0.0275078 to 0.0328455; mean 0.0306	0.0328455 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	0.000197921 to 0.000318768; mean 0.0003	0.000306598 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	4 / 4	4.01789e-05 to 5.22023e-05; mean 0	4.15036e-05 at 2026-08-02 19:10Z; 1 h 10 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-08-02 19:10Z; 1 h 10 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-6.07222e-05 to 0.000178122; mean 0	-6.07222e-05 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
tceq	Carbon monoxide (co)	ppm	3 / 188	0 to 0.7; mean 0.1957	0.2 at 2026-08-02 17:00Z; 1 h before event
tceq	Nitrogen dioxide (no2)	ppb	12 / 735	-2.4 to 33.6; mean 4.5578	2.3 at 2026-08-02 18:00Z; at event time
tceq	Sulfur dioxide (so2)	ppb	3 / 187	-0.6 to 7.2; mean -0.0214	-0.2 at 2026-08-02 18:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	08-01 06Z 92.54, 12Z 74.28, 18Z 49.66, 08-02 00Z 76.74, 06Z 87.55, 12Z 63.03, 18Z 37.03, 08-03 00Z 71.02, 06Z 89.41, 12Z 69.23, 18Z 46.67, 08-04 00Z 76.64
asos	Air temperature (temperature)	08-01 06Z 25.73, 12Z 30.27, 18Z 34.17, 08-02 00Z 29.94, 06Z 27.81, 12Z 30.98, 18Z 35.04, 08-03 00Z 29.45, 06Z 26.19, 12Z 30.55, 18Z 34.3, 08-04 00Z 28.38
asos	Wind direction (wind_direction)	08-01 06Z 164.9, 12Z 232.8, 18Z 229.2, 08-02 00Z 205.5, 06Z 232.6, 12Z 138.6, 18Z 186.6, 08-03 00Z 166.7, 06Z 134.2, 12Z 149.5, 18Z 158.3, 08-04 00Z 157.4
asos	Wind speed (wind_speed)	08-01 06Z 2.126, 12Z 4.089, 18Z 3.175, 08-02 00Z 3.187, 06Z 2.847, 12Z 1.827, 18Z 2.893, 08-03 00Z 3.04, 06Z 1.523, 12Z 2.163, 18Z 4.523, 08-04 00Z 3.447
noaa_gfs	500-hPa geopotential height (gh_500)	08-01 06Z 5945, 12Z 5927, 18Z 5927, 08-02 00Z 5905, 06Z 5902, 12Z 5882, 18Z 5887, 08-03 00Z 5877, 06Z 5888, 12Z 5880, 18Z 5896, 08-04 00Z 5899, 06Z 5914
noaa_gfs	Planetary boundary layer height (pbl_height)	08-01 06Z 378.1, 12Z 334.6, 18Z 1620, 08-02 00Z 868, 06Z 297.3, 12Z 185.6, 18Z 2111, 08-03 00Z 987.8, 06Z 171.5, 12Z 101.6, 18Z 2227, 08-04 00Z 954.5, 06Z 279
noaa_gfs	Precipitable water (precipitable_water)	08-01 06Z 37.74, 12Z 42.75, 18Z 48.44, 08-02 00Z 56.82, 06Z 53.55, 12Z 42.2, 18Z 33.29, 08-03 00Z 30.71, 06Z 32.5, 12Z 33.01, 18Z 34.71, 08-04 00Z 39.41, 06Z 33.65
noaa_gfs	Surface pressure (surface_pressure)	08-01 06Z 1012, 12Z 1012, 18Z 1010, 08-02 00Z 1006, 06Z 1008, 12Z 1009, 18Z 1009, 08-03 00Z 1006, 06Z 1009, 12Z 1009, 18Z 1009, 08-04 00Z 1008, 06Z 1010
noaa_gfs	850-hPa temperature (t_850)	08-01 06Z 22.23, 12Z 21.82, 18Z 19.7, 08-02 00Z 21.55, 06Z 20.62, 12Z 20.6, 18Z 19.51, 08-03 00Z 20.75, 06Z 19.31, 12Z 18.56, 18Z 18.92, 08-04 00Z 19.8, 06Z 20.04
noaa_gfs	10-m eastward wind (u_10m)	08-01 06Z 1.506, 12Z 2.167, 18Z 4.049, 08-02 00Z 1.812, 06Z 3.034, 12Z 1.94, 18Z -0.5041, 08-03 00Z -1.151, 06Z 1.25, 12Z 1.441, 18Z -0.8189, 08-04 00Z -1.949, 06Z 0.6604
noaa_gfs	10-m northward wind (v_10m)	08-01 06Z 3.205, 12Z 1.839, 18Z 0.4296, 08-02 00Z 0.163, 06Z 1.85, 12Z -1.332, 18Z -2.868, 08-03 00Z -0.3577, 06Z 2.523, 12Z 1.111, 18Z 1.813, 08-04 00Z 4.831, 06Z 2.869
openaq	Ozone (ozone)	08-01 06Z 0.01177, 12Z 0.01407, 18Z 0.03085, 08-02 00Z 0.02182, 06Z 0.009148, 12Z 0.0239, 18Z 0.05994, 08-03 00Z 0.05262, 06Z 0.02483, 12Z 0.0207, 18Z 0.05074, 08-04 00Z 0.0222, 06Z 0.016
openaq	PM10 (pm10)	08-01 06Z 16.94, 12Z 30.22, 18Z 31.83, 08-02 00Z 24.33, 06Z 22.22, 12Z 22.89, 18Z 16.61, 08-03 00Z 20.44, 06Z 20.5, 12Z 50.18, 18Z 43.35, 08-04 00Z 41.67, 06Z 38.33
openaq	PM2.5 (pm25)	08-01 06Z 3.385, 12Z 4.946, 18Z 5.681, 08-02 00Z 8.976, 06Z 9.961, 12Z 8.137, 18Z 8.889, 08-03 00Z 10.5, 06Z 9.138, 12Z 9.965, 18Z 12.13, 08-04 00Z 11.81, 06Z 15.16
openweather	Cloud cover (cloud_cover)	08-01 06Z 5.417, 12Z 15, 18Z 14.68, 08-02 00Z 59.04, 06Z 67.04, 12Z 90.6, 18Z 38.17, 08-03 00Z 24.62, 06Z 1.348, 12Z 0, 18Z 2.625, 08-04 00Z 20.54
openweather	Relative humidity (humidity)	08-01 06Z 87.79, 12Z 81.52, 18Z 56.28, 08-02 00Z 72.92, 06Z 86.22, 12Z 72.44, 18Z 42.25, 08-03 00Z 66.92, 06Z 83.7, 12Z 76.32, 18Z 51.58, 08-04 00Z 72.88
openweather	Precipitation rate (precipitation)	08-01 06Z 0, 12Z 0, 18Z 0.0048, 08-02 00Z 0, 06Z 0.009565, 12Z 0, 18Z 0, 08-03 00Z 0, 06Z 0, 12Z 0, 18Z 0, 08-04 00Z 0
openweather	Surface pressure (pressure)	08-01 06Z 1013, 12Z 1014, 18Z 1010, 08-02 00Z 1009, 06Z 1010, 12Z 1012, 18Z 1009, 08-03 00Z 1009, 06Z 1011, 12Z 1012, 18Z 1010, 08-04 00Z 1011
openweather	Air temperature (temperature)	08-01 06Z 26.65, 12Z 29.63, 18Z 35.69, 08-02 00Z 31.3, 06Z 28.36, 12Z 30.7, 18Z 36.38, 08-03 00Z 31.26, 06Z 27.19, 12Z 30.29, 18Z 35.86, 08-04 00Z 30
openweather	Wind direction (wind_direction)	08-01 06Z 185.4, 12Z 238, 18Z 266.2, 08-02 00Z 223.9, 06Z 234.9, 12Z 92.44, 18Z 161.8, 08-03 00Z 168.4, 06Z 211.5, 12Z 214.4, 18Z 153.9, 08-04 00Z 179.9
openweather	Wind speed (wind_speed)	08-01 06Z 2.443, 12Z 2.603, 18Z 2.076, 08-02 00Z 2.311, 06Z 2.09, 12Z 1.979, 18Z 2.005, 08-03 00Z 2.236, 06Z 2.183, 12Z 1.3, 18Z 2.382, 08-04 00Z 2.62
purpleair	PM2.5 (pm25)	08-01 06Z 4.459, 12Z 6.313, 18Z 7.207, 08-02 00Z 12.71, 06Z 12.21, 12Z 10.38, 18Z 9.468, 08-03 00Z 12.45, 06Z 12.11, 12Z 12.36, 18Z 13.92, 08-04 00Z 13.2
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1

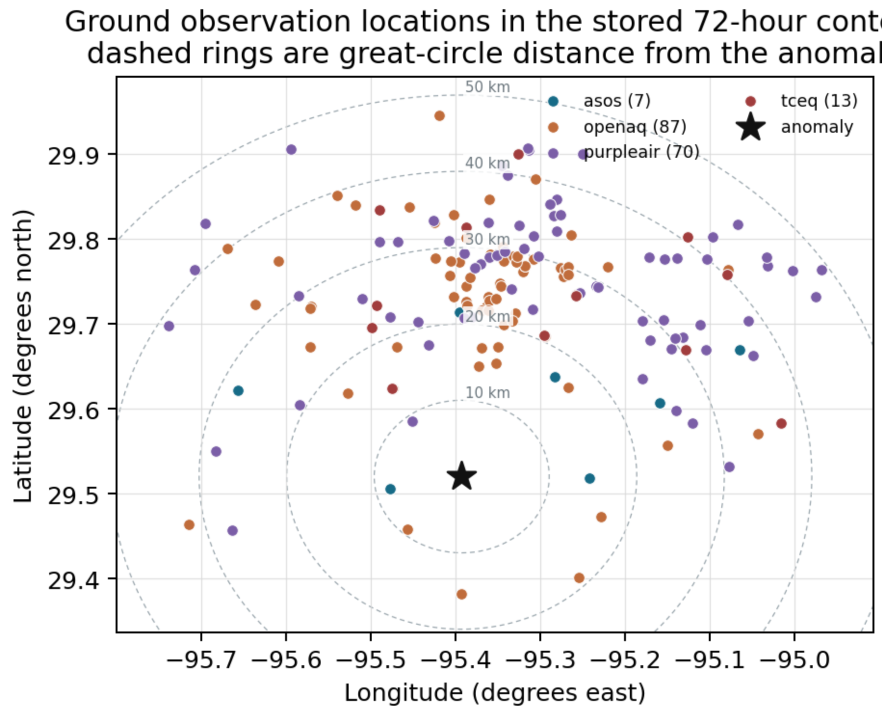
Source	Metric	Values
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	CO column (s5p_co_column)	08-01 18Z 0.02751, 08-02 18Z 0.03284, 08-03 18Z 0.03095
sentinel5p	CO granules available (s5p_co_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	08-01 18Z 0.0001979, 08-02 18Z 0.0003127, 08-03 18Z 0.0002755
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	08-01 18Z 4.018e-05, 08-02 18Z 4.15e-05, 08-03 18Z 4.973e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	08-01 18Z 6.519e-06, 08-02 18Z -4.61e-05, 08-03 18Z 0.0001056
sentinel5p	SO2 granules available (s5p_so2_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
tceq	Carbon monoxide (co)	08-01 06Z 0.1611, 12Z 0.2056, 18Z 0.1812, 08-02 00Z 0.22, 06Z 0.1167, 12Z 0.1333, 18Z 0.16, 08-03 00Z 0.2444, 06Z 0.2389, 12Z 0.2722, 18Z 0.25, 08-04 00Z 0.2111, 06Z 0.1333
tceq	Nitrogen dioxide (no2)	08-01 06Z 3.42, 12Z 3.552, 18Z 4.337, 08-02 00Z 6.082, 06Z 3.036, 12Z 3.271, 18Z 2.811, 08-03 00Z 4.778, 06Z 8.049, 12Z 7.837, 18Z 4.965, 08-04 00Z 2.549, 06Z 2.683
tceq	Sulfur dioxide (so2)	08-01 06Z -0.1944, 12Z -0.1167, 18Z -0.04706, 08-02 00Z -0.18, 06Z -0.07778, 12Z -0.09444, 18Z -0.02778, 08-03 00Z -0.03333, 06Z -0.2056, 12Z 0.7444, 18Z -0.07222, 08-04 00Z -0.04444, 06Z -0.2667

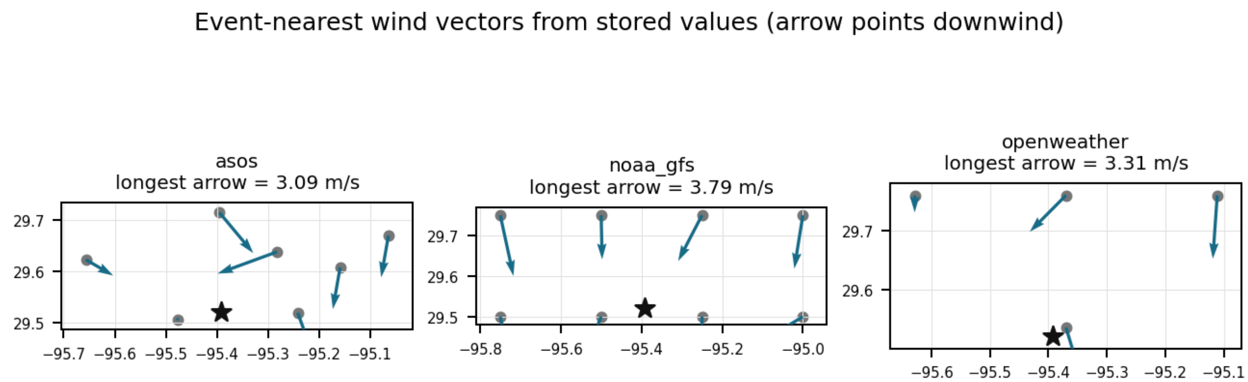
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	Relative humidity (humidity)	AXH (8.318 km) 68.99, LVJ (14.592 km) 71.43, HOU (16.828 km) 69.1, MCJ (21.54 km) 66.34, EFD (24.595 km) 67.37, SGR (27.938 km) 73.47, T41 (35.81 km) 70.28
asos	Air temperature (temperature)	AXH (8.318 km) 29.95, LVJ (14.592 km) 30.14, HOU (16.828 km) 30.54, MCJ (21.54 km) 30.58, EFD (24.595 km) 30.84, SGR (27.938 km) 29.88, T41 (35.81 km) 30.17
asos	Wind direction (wind_direction)	AXH (8.318 km) 107.2, LVJ (14.592 km) 163.9, HOU (16.828 km) 205.2, MCJ (21.54 km) 204.4, EFD (24.595 km) 174.5, SGR (27.938 km) 207, T41 (35.81 km) 196.8
asos	Wind speed (wind_speed)	AXH (8.318 km) 1.133, LVJ (14.592 km) 2.19, HOU (16.828 km) 3.569, MCJ (21.54 km) 3.651, EFD (24.595 km) 4.29, SGR (27.938 km) 2.897, T41 (35.81 km) 3.454
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.50,-95.50 (10.645 km) 5903, gfs:29.50,-95.25 (13.973 km) 5902, gfs:29.75,-95.50 (27.574 km) 5902, gfs:29.75,-95.25 (29.018 km) 5902, gfs:29.50,-95.75 (34.669 km) 5903, gfs:29.50,-95.00 (38.049 km) 5901, gfs:29.75,-95.75 (42.968 km) 5903, gfs:29.75,-95.00 (45.732 km) 5901
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.50,-95.50 (10.645 km) 760, gfs:29.50,-95.25 (13.973 km) 703.5, gfs:29.75,-95.50 (27.574 km) 1085, gfs:29.75,-95.25 (29.018 km) 1009, gfs:29.50,-95.75 (34.669 km) 752.8, gfs:29.50,-95.00 (38.049 km) 709.9, gfs:29.75,-95.75 (42.968 km) 653.2, gfs:29.75,-95.00 (45.732 km) 796.7
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.50,-95.50 (10.645 km) 40.18, gfs:29.50,-95.25 (13.973 km) 40.62, gfs:29.75,-95.50 (27.574 km) 39.54, gfs:29.75,-95.25 (29.018 km) 39.98, gfs:29.50,-95.75 (34.669 km) 39.79, gfs:29.50,-95.00 (38.049 km) 40.47, gfs:29.75,-95.75 (42.968 km) 38.85, gfs:29.75,-95.00 (45.732 km) 39.83
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.50,-95.50 (10.645 km) 1009, gfs:29.50,-95.25 (13.973 km) 1010, gfs:29.75,-95.50 (27.574 km) 1008, gfs:29.75,-95.25 (29.018 km) 1009, gfs:29.50,-95.75 (34.669 km) 1008, gfs:29.50,-95.00 (38.049 km) 1011, gfs:29.75,-95.75 (42.968 km) 1007, gfs:29.75,-95.00 (45.732 km) 1010

Source	Metric	Values
noaa_gfs	850-hPa temperature (t_850)	gfs:29.50,-95.50 (10.645 km) 20.34, gfs:29.50,-95.25 (13.973 km) 20.21, gfs:29.75,-95.50 (27.574 km) 20.3, gfs:29.75,-95.25 (29.018 km) 20.16, gfs:29.50,-95.75 (34.669 km) 20.45, gfs:29.50,-95.00 (38.049 km) 20.13, gfs:29.75,-95.75 (42.968 km) 20.42, gfs:29.75,-95.00 (45.732 km) 20.09
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.50,-95.50 (10.645 km) 0.9337, gfs:29.50,-95.25 (13.973 km) 0.9283, gfs:29.75,-95.50 (27.574 km) 1.098, gfs:29.75,-95.25 (29.018 km) 0.7845, gfs:29.50,-95.75 (34.669 km) 1.384, gfs:29.50,-95.00 (38.049 km) 1.083, gfs:29.75,-95.75 (42.968 km) 1.306, gfs:29.75,-95.00 (45.732 km) 0.7514
noaa_gfs	10-m northward wind (v_10m)	gfs:29.50,-95.50 (10.645 km) 1.345, gfs:29.50,-95.25 (13.973 km) 1.656, gfs:29.75,-95.50 (27.574 km) 1.067, gfs:29.75,-95.25 (29.018 km) 1.06, gfs:29.50,-95.75 (34.669 km) 0.9803, gfs:29.50,-95.00 (38.049 km) 2.181, gfs:29.75,-95.75 (42.968 km) 0.788, gfs:29.75,-95.00 (45.732 km) 0.815
openaq	Ozone (ozone)	3055 (0.0 km) 0.02859, 312 (13.969 km) 0.0303, 320 (16.85 km) 0.0269, 325 (20.747 km) 0.0125, 311 (22.073 km) 0.03034, 3378 (27.1 km) 0.02743, 1319333 (28.091 km) 0.03497, 332 (30.475 km) 0.03184, +8 more
openaq	PM10 (pm10)	13380082 (27.1 km) 41.51, 15852419 (28.582 km) 18.48, 271 (30.475 km) 25.13
openaq	PM2.5 (pm25)	14152710 (9.263 km) 13.05, 8967021 (14.63 km) 8.895, 15472101 (15.293 km) 5.806, 11987924 (15.404 km) 7.483, 11638043 (16.705 km) 7.106, 14157975 (16.848 km) 8.496, 9416627 (16.987 km) 5.518, 15399615 (16.994 km) 7.362, +60 more
openweather	Cloud cover (cloud_cover)	grid:south (2.792 km) 31.94, grid:center (26.788 km) 26.07, grid:west (35.114 km) 28.04, grid:east (38.11 km) 27.28
openweather	Relative humidity (humidity)	grid:south (2.792 km) 70.89, grid:center (26.788 km) 68.6, grid:west (35.114 km) 69.46, grid:east (38.11 km) 74.01
openweather	Precipitation rate (precipitation)	grid:south (2.792 km) 0, grid:center (26.788 km) 0, grid:west (35.114 km) 0.001667, grid:east (38.11 km) 0.003056
openweather	Surface pressure (pressure)	grid:south (2.792 km) 1011, grid:center (26.788 km) 1011, grid:west (35.114 km) 1011, grid:east (38.11 km) 1011
openweather	Air temperature (temperature)	grid:south (2.792 km) 30.95, grid:center (26.788 km) 31.65, grid:west (35.114 km) 31.14, grid:east (38.11 km) 30.86
openweather	Wind direction (wind_direction)	grid:south (2.792 km) 213.3, grid:center (26.788 km) 200, grid:west (35.114 km) 189.7, grid:east (38.11 km) 172.3
openweather	Wind speed (wind_speed)	grid:south (2.792 km) 2.159, grid:center (26.788 km) 2.077, grid:west (35.114 km) 1.383, grid:east (38.11 km) 3.103
purpleair	PM2.5 (pm25)	161003 (9.186 km) 5.34, 166451 (17.628 km) 10.42, 293735 (20.756 km) 12.05, 276512 (20.799 km) 11.63, 46237 (20.895 km) 0.3681, 293725 (20.922 km) 12.37, 186315 (22.489 km) 11.82, 6752 (23.328 km) 7.511, +62 more
tceq	Carbon monoxide (co)	482011035 (27.078 km) 0.1667, 482011039 (30.456 km) 0.1475, 482011052 (32.704 km) 0.2703
tceq	Nitrogen dioxide (no2)	480391004 (0.017 km) 1.683, 482010416 (20.738 km) 4.703, 482010055 (22.069 km) 2.61, 482011066 (24.389 km) 7.859, 482011035 (27.078 km) 7.105, 482011039 (30.456 km) 3.05, 482011052 (32.704 km) 9.068, 482010047 (36.131 km) 3.44, +4 more
tceq	Sulfur dioxide (so2)	482010051 (13.984 km) -0.2492, 482011035 (27.078 km) -0.2825, 482011039 (30.456 km) 0.4836


Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly
50 km
40 km
30 km
20 km
10 km
asos (7)
openaq (87)
purpleair (70)
tceq (13)
anomaly
Latitude (degrees north)
29.9
29.8
29.7
29.6
29.5
29.4
−95.7 −95.6 −95.5 −95.4 −95.3 −95.2 −95.1 −95.0
Longitude (degrees east)


Event-nearest wind vectors from stored values (arrow points downwind)
asos
longest arrow = 3.09 m/s
29.7
29.6
29.5
−95.7 −95.6 −95.5 −95.4 −95.3 −95.2 −95.1
noaa_gfs
longest arrow = 3.79 m/s
29.7
29.6
29.5
−95.8 −95.6 −95.4 −95.2 −95.0
openweather
longest arrow = 3.31 m/s
29.7
29.6
−95.6 −95.5 −95.4 −95.3 −95.2 −95.1

Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

Hot and dry afternoon meteorology strongly supports enhanced local photochemical ozone production in central Houston, because surface temperature near the anomaly reached about 35 °C while relative humidity fell to roughly 43–47% and regional precipitable water dropped to about 35 mm during the event window.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

The air quality anomaly in Houston, Texas is caused by a combination of high temperatures and wind direction.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

The air quality anomaly in Houston, Texas is characterized by elevated levels of PM2.5 and NO2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

The PM2.5 levels are elevated at 12.5 ug/m3, which is above the mean of 10.5843 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

Weak and variable near-surface winds support stagnation or recirculation as a contributor to the ozone spike, because the nearest ASOS station reported calm wind at 17:55 UTC, area 10 m winds in gfs were only about 3 m/s with changing flow direction, and surface observations showed inconsistent directions around the anomaly time rather than a strong flushing regime.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

Hot, unusually dry afternoon conditions favored rapid photochemical ozone production in Houston at the time of the anomaly, as surface temperature was about 35 to 35.3 C while humidity fell to about 42 to 47 percent near 18Z on 2026-08-02.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

Elevated atmospheric formaldehyde column levels reaching 0.0003127 mol/m² on August 2, 2026, indicate abundant volatile organic compound precursors that supported rapid photochemical ozone synthesis.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

The temperature and wind direction in the affected area were within normal ranges on August 2nd, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

The temperature was unusually high on August 2nd, leading to increased emissions from industrial sources and vehicles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

A strong wind pattern on August 2nd brought pollutants from nearby cities into the Houston area, contributing to the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

Surface ozone concentrations reached a severe anomalous peak of 0.075 ppm in Houston at 18:00 UTC on August 2, 2026, during high afternoon temperatures reaching 35.0 °C to 36.4 °C.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

Elevated atmospheric formaldehyde (HCHO) column densities observed around 18:38 UTC on August 2, 2026, indicate increased availability of volatile organic compound (VOC) precursors for photochemical ozone synthesis.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

The combination of high temperatures and stagnant air patterns on August 2nd led to a buildup of pollutants in the Houston area, causing the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

The openaq ozone network within 50 km had a 6-hour mean ozone concentration of 0.05994 ppm for 2026-08-02 18Z, compared with 0.0239 ppm for 2026-08-02 12Z.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

Sentinel-5P satellite retrievals recorded a peak formaldehyde column concentration of 0.0003127 mol/m^2 on August 2, 2026, indicating elevated precursor volatile organic compound levels that fueled ozone production.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

High ambient surface temperatures reaching up to 37°C and decreased cloud cover during afternoon hours on August 2, 2026, accelerated photochemical reactions responsible for ground-level ozone formation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

Houston co-pollutant observations on 2026-08-02 are more consistent with VOC-rich urban-industrial ozone chemistry than with a fresh combustion, sulfur, or smoke event because TCEQ NO2 and SO2 near the anomaly were low, PM2.5 and PM10 were not strongly elevated, Sentinel-5P HCHO and CO were elevated while Sentinel-5P SO2 was not elevated, and Sentinel-5P NO2 was not unusually high.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

The air quality anomaly in Houston, Texas is not caused by CO levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

Surface weather observations from ASOS and OpenWeather at 18:00 UTC on August 2, 2026, indicated ambient temperatures exceeding 35 degrees Celsius and relative humidity levels between 37% and 42%.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

The openaq ozone anomaly at 29.520, -95.392 had an expected value of 0.017044303797468352 ppm and a z-score of 5.762711911484322 at 2026-08-02T18:00:00+00:00, which is labeled severe in the provided data.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

There is no evidence of any significant industrial or agricultural activities that could have contributed to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

Weak and variable near-surface winds likely promoted stagnation or short-range recirculation that allowed ozone and its precursors to accumulate over central Houston near the time of the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

Houston surface ozone on 2026-08-02 18:00Z is most consistent with a daytime photochemical production event because area ozone increased sharply to a 6-hour mean near 0.05994 ppm while surface temperature was about 35 to 36 C and humidity fell to roughly 37 to 47 percent.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

The precursor and co-pollutant pattern supports VOC-rich urban-industrial photochemistry more than a fresh combustion plume or smoke/dust event, because ozone was high while local NO2 remained low near 2.3 ppb, PM10 and PM2.5 were not elevated, satellite HCHO and CO were somewhat elevated, and satellite SO2 showed no corresponding enhancement.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

Ambient surface temperatures exceeding 35 degrees Celsius alongside relative humidity dropping below 40 percent around 18:00 UTC on August 2, 2026, created optimal photochemical conditions for rapid ozone formation in Houston.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

The precursor signature is most consistent with VOC-driven urban-industrial photochemistry rather than a fresh NOx or SO2 plume, because ozone and formaldehyde were elevated while nearby NO2, SO2, and CO were not elevated at the anomaly time.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

Sentinel-5P satellite measurements at approximately 18:38 UTC on August 2, 2026, observed an elevated tropospheric formaldehyde column density of approximately 0.000313 mol/m2 across the regional domain.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

Stagnant surface winds averaging under 3 m/s and localized calm wind conditions restricted regional pollutant ventilation, promoting surface ozone accumulation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

ASOS surface wind observations near the anomaly site recorded near-calm winds of 0.0 m/s around 17:55 UTC on August 2, 2026, favoring localized precursor stagnation prior to peak ozone generation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

The openaq monitor at 29.520, -95.392 recorded an ozone concentration of 0.075 ppm at 2026-08-02T18:00:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

Houston meteorology on 2026-08-02 18:00Z favored ozone accumulation and precursor processing because local winds were weak or variable, one nearby ASOS observation reported calm wind, and GFS showed a deep afternoon boundary layer near 2408 m under relatively low precipitable water near 34.9 mm.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

Weak surface wind speeds below 3 m/s combined with deep planetary boundary layer growth on August 2, 2026, created local atmospheric stagnation that facilitated ozone accumulation near the surface.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

OpenAQ recorded a severe ozone anomaly reaching 0.075 ppm against an expected value of 0.017 ppm at coordinates (29.520, -95.392) on August 2, 2026, at 18:00 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

The anomaly occurred on August 2nd at around 18:01 UTC, with a nearest sensor located approximately 24.318 km away from the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.